Mandatory Excursions in Privacy-Constrained Probability Paths

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We study positive probability paths whose coordinate log derivatives are bounded by a privacy budget. For every swapped triplet (c, a, b) and (c, b, a), with $0 < a < b$ and $c = 1 - a - b > 0$, we determine the minimum feasible budget T_* and the smallest possible peak of the unchanged-endpoint coordinate at every budget $L \geq T_*$. The peak is $\max\{c, 1 - 2ae^{L/2}\}$, attained by an explicit path; the minimum-budget path is unique. We also determine the sharp cost of requiring every coordinate to be monotone, or to remain between its endpoint values. At endpoint privacy distance $\eta > 0$, the worst ratio to the unrestricted optimum is $2\log((e^\eta + 1)/2)/\eta$ in every dimension at least three, with positive attaining examples, whereas the binary ratio is one. For general endpoints we give necessary excursion certificates evaluable exactly by a rational breakpoint sweep and one square root per coordinate.

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1 Introduction

A mechanism on an interval assigns a probability vector to each input. For a finite output alphabet, bounding every coordinate log derivative by L guarantees both directions of metric differential privacy between every pair of inputs. Normalization couples the coordinates: keeping an output probability between its endpoint values can require a larger budget than allowing it to rise temporarily. We quantify this effect exactly for a three-output family and identify the sharp worst cost of preserving coordinate shape in any dimension.

For $n \geq 2$, let $\Delta_n^\circ = \{x \in (0,1)^n : \sum_i x_i = 1\}$. All logarithms are natural. For positive endpoints $p, q \in \Delta_n^\circ$, write

$$d_{\mathrm{DP}}(p,q) = \max_i \left|\log \frac{p_i}{q_i}\right|, \qquad \mathcal{L}(r) = \operatorname*{ess\,sup}_{t\in[0,1]} \max_i \frac{|r_i'(t)|}{r_i(t)}. \tag{1}$$

Paths $r : [0,1] \to \Delta_n^\circ$ are absolutely continuous, with $r(0) = p$ and $r(1) = q$. Define $\Lambda(p,q) = \inf_r \mathcal{L}(r)$. A finite budget $\mathcal{L}(r) \leq L$ implies $d_{\mathrm{DP}}(r(s), r(t)) \leq L|s-t|$ by integration, so it controls the whole interval, including its endpoints.

Let $\Lambda_{\mathrm{mon}}(p,q)$ restrict the infimum to paths whose coordinates are monotone in their endpoint directions, with equal endpoints forcing a constant coordinate. Let $\Lambda_{\mathrm{box}}(p,q)$ instead require

$$\min\{p_i, q_i\} \leq r_i(t) \leq \max\{p_i, q_i\} \qquad \text{for all } i, t. \tag{2}$$

The affine path belongs to both classes, and $\Lambda \leq \Lambda_{\mathrm{box}} \leq \Lambda_{\mathrm{mon}}$. Our main result determines the entire tradeoff between budget and peak probability for swapped triplets. It then yields the exact worst ratios $\Lambda_{\mathrm{mon}}/\Lambda$ and $\Lambda_{\mathrm{box}}/\Lambda$. A final theorem evaluates general necessary peak certificates in exact arithmetic.

Relation to existing results. Borgs, Chayes, Smith and Zadik [BCSZ18], Proposition 2.1, prove that a private algorithm on a subset of an arbitrary metric input space admits an extension with twice the privacy parameter. This is the relevant general extension baseline. Dong, Roth and Su [DRS19], Theorem 2.10 and Proposition 2.15, provide Blackwell comparison and the continuous-group Laplace limit; these are background for unrestricted reachability. Our proofs use elementary scalar speed bounds to study additional constraints on the entire normalized path.

Qiu [Qiu26], Definitions 5–6 and Proposition 3, explicitly normalizes log-convex interpolation and accounts for its privacy cost. Liu and Qiu [LQ26], Section 5.2 and Appendix E, calibrate pre-normalization budgets and use endpoint envelopes in their graph-based extension framework. The optimization here minimizes an individual output's peak over *all* admissible normalized paths. Normalization can move coordinates outside their endpoint ranges, so the restriction in (2) must be checked on the normalized probabilities themselves.

The endpoint metric in (1) is the restriction of the positive-orthant Thompson metric. Lemmens and Roelands [LR15], Theorems 4.3 and 5.1, study uniqueness for cone Thompson geodesics; their Theorem 5.2 concerns Hilbert geodesics on a section. We prove the stated normalized-path uniqueness directly by saturation. Fernández de Bobadilla, Hontarenko, Müller-Rigat and Życzkowski [FHMRŻ26], Proposition 6 and Theorem 8, study a one-sided instantaneous cost $F_\infty(p,v) = \max_{v_i<0} |v_i|/p_i$ (with value zero at $v = 0$) and its optimized endpoint quasi-distance. Our instantaneous cost is $\max\{F_\infty(p,v), F_\infty(p,-v)\}$ before optimizing paths. The two optimizations differ. The contribution here is the all-budget excursion frontier and the sharp cost of coordinate restrictions; the scalar binary calculation, affine length argument and endpoint-envelope principle are standard ingredients.

2 An exact budget–peak frontier

Index the coordinates of a triplet by $0, 1, 2$. Coordinate 0, whose endpoints agree, will be called the reservoir. For fixed endpoints define

$$P_0(L) = \inf_{r:\, \mathcal{L}(r)\leq L} \max_{t\in[0,1]} r_0(t).$$

The next theorem includes an attaining path, so the feasible infima are minima.

THEOREM 2.1 (SWAPPED-TRIPLET FRONTIER). *Let $0 < a < b$, $a + b < 1$, $c = 1 - a - b$, $p = (c, a, b)$ and $q = (c, b, a)$. Put*

$$T_* = \begin{cases} \log(b/a), & b \le \frac{1}{2}, \\ \log\big(1/[4a(1-b)]\big), & b > \frac{1}{2}, \end{cases} \qquad T_0 = 2\log\frac{a+b}{2a}. \tag{3}$$

A path of budget at most L exists exactly when $L \ge T_*$. For every such L,

$$P_0(L) = \max\{c,\ 1 - 2ae^{L/2}\}. \tag{4}$$

In particular $\Lambda(p, q) = T_* < T_0$. The unit-interval path with budget at most T_* is unique. Its reservoir peak is $1 - 2\sqrt{ab}$ when $b \le 1/2$, and $1 - \sqrt{a/(1-b)}$ when $b > 1/2$; both exceed c.

Thus the optimal budget forces an excursion of an output with identical endpoint probabilities. Increasing the budget reduces its minimum peak continuously, until the reservoir can remain constant at $L = T_0$. Table 1 summarizes the two parameter regimes.

Table 1. Swapped-triplet thresholds and the peak at minimum budget. In both regimes $P_0(L) = 1 - 2ae^{L/2}$ for $T_* \le L \le T_0$ and $P_0(L) = c$ for $L \ge T_0$.

Regime	Minimum budget T_*	Peak $P_0(T_*)$
$b \le 1/2$	$\log(b/a)$	$1 - 2\sqrt{ab}$
$b > 1/2$	$\log(1/[4a(1-b)])$	$1 - \sqrt{a/(1-b)}$

2.1 Scalar speed bounds

If $\mathcal{L}(r) \le L$, integrating each log derivative from both endpoints gives

$$r_i(t) \le \min\{p_i e^{Lt}, q_i e^{L(1-t)}\}. \tag{5}$$

Each positive coordinate is bounded away from zero on the compact interval, so its logarithm is absolutely continuous and this integration is valid. Differentiating normalization almost everywhere also gives

$$|r_i'| \le L r_i, \qquad |r_i'| = \left|\sum_{k \ne i} r_k'\right| \le L(1 - r_i). \tag{6}$$

Define the increasing scalar transform

$$\Phi(x) = \begin{cases} \log(2x), & 0 < x \le 1/2, \\ -\log(2(1-x)), & 1/2 \le x < 1. \end{cases} \tag{7}$$

It has derivative $1/\min\{x, 1-x\}$, including at $1/2$. Hence $|(\Phi(r_i))'| \le L$ almost everywhere. For binary distributions $(x, 1-x)$ and $(y, 1-y)$ the minimum budget is therefore

$$|\Phi(y) - \Phi(x)|. \tag{8}$$

The path $x(t) = \Phi^{-1}((1-t)\Phi(x) + t\Phi(y))$ attains this value and is monotone. This elementary calculation will also determine the restricted optima.

2.2 Proof of Theorem 2.1

PROOF. The changing coordinate from a to b and (8) force $L \ge T_*$. At $t = 1/2$, equation (5) bounds each changing coordinate by $ae^{L/2}$. Normalization gives $r_0(1/2) \ge 1 - 2ae^{L/2}$, while the endpoints force a peak at least c. These prove the lower bounds.

First observe that $T_* < T_0$. Indeed,

$$T_* = \int_a^b \frac{dx}{\min\{x, 1-x\}}, \qquad T_0 = \int_a^b \frac{dx}{\min\{x, a+b-x\}}.$$

The first denominator is at least the second throughout, and strictly greater on $((a+b)/2, b)$, since $a+b<1$.

For $T_* \le L \le T_0$, put $B = e^{L/2}$ and $H = 1-2aB$. Then $H \ge c > 0$ and $aB \le (a+b)/2 < 1/2$. On the first half of the interval use physical time $s = Lt$ and $u = e^s$, so $1 \le u \le B$. Let g be the fastest binary decrease from b at unit physical-time budget:

$$g(u) = \begin{cases} b/u, & b \le 1/2, \\ 1-(1-b)u, & b > 1/2,\ 1 \le u \le \kappa, \\ K/u, & b > 1/2,\ u \ge \kappa, \end{cases} \qquad \kappa = \frac{1}{2(1-b)}, \quad K = \frac{1}{4(1-b)} \tag{9}$$

in the last two cases. When $b \le 1/2$, set $K = b$. Define $h(u) = 1 - au - g(u)$ and

$$r_1 = au, \qquad r_0 = \min\{H, h(u)\}, \qquad r_2 = \max\{g(u), 2aB - au\}. \tag{10}$$

We verify that this is an admissible half-path with reservoir peak H.

Let $u_* = \sqrt{K/a}$. The budget condition implies $B \ge u_*$. For $b > 1/2$ we also have $u_* > \kappa$, since $a < 1-b$. In this case $h(u) = cu$ on $[1, \kappa]$; on its later piece, and throughout when $b \le 1/2$, it equals $1 - au - K/u$. Thus h increases from c to its maximum at u_*, and decreases thereafter. At the last point,

$$h(B) - H = aB - K/B \ge 0.$$

It follows that $h \ge c$ on $[1, B]$ and that h first reaches the cap H at some u_0, remaining at least H through B. When $H = c$ take $u_0 = 1$. Formula (10) consequently consists of an initial uncapped part followed by a constant-reservoir part. Its coordinates are positive and sum to one. At $u = B$ their values are (H, aB, aB).

A dot in the next checks denotes differentiation in s. In the initial uncapped binary phase when $b > 1/2$, the vector is

$$(r_0, r_1, r_2) = (cu, au, 1-(1-b)u).$$

The first two derivatives equal their coordinates, and $|\dot r_2| = 1 - r_2 \le r_2$ because $r_2 \ge 1/2$. In any later uncapped phase, $\dot r_1 = r_1$, $\dot r_2 = -r_2$ and $\dot r_0 = r_2 - r_1$. Both changing coordinates are then at most $1/2$, so

$$r_0 + \dot r_0 = 1 - 2r_1 \ge 0, \qquad r_0 - \dot r_0 = 1 - 2r_2 \ge 0.$$

On a capped part, $\dot r_0 = 0$ and $\dot r_2 = -r_1$. Here $r_2 = 2aB - au \ge au = r_1$, because $u \le B$. These relations give $|\dot r_i| \le r_i$ on every piece. The formulas agree at their finitely many joins and are absolutely continuous. Reflect the half-path in time and exchange coordinates $1, 2$ to define the second half. Its midpoint matches, its terminal value is q, and scaling $s = Lt$ proves $\mathcal{L}(r) \le L$. The peak of r_0 is exactly H.

At $L = T_0$ the cap is c, yielding the constant-reservoir binary path. For $L > T_0$ reuse this unit-interval path, of cost T_0. Its peak c attains the lower bound. This proves feasibility and (4) at every budget.

Finally suppose $\mathcal{L}(r) \le T_*$. Then $\Phi(r_1(1)) - \Phi(r_1(0)) = T_*$ and $|(\Phi(r_1))'| \le T_*$ almost everywhere. The nonnegative deficit has integral zero, forcing $(\Phi(r_1))' = T_*$ almost everywhere. Similarly $(\Phi(r_2))' = -T_*$. Consequently

$$r_1(t) = \Phi^{-1}(\Phi(a) + T_* t), \qquad r_2(t) = \Phi^{-1}(\Phi(b) - T_* t),$$

and normalization fixes r_0. The construction proves this forced path exists. Substitution in (4) gives the stated peaks, each strictly above c because $T_* < T_0$. □

For example, $a = 1/16$, $b = 3/4$ and $c = 3/16$ give endpoint distance $\log 12$ and minimum budget $\log 16$. Every minimum-budget path reaches reservoir probability $1/2$, while a constant reservoir requires $T_0 = 2\log(13/2)$. This illustrates why the two endpoint privacy inequalities alone do not determine a feasible budget for the whole interval.

3 The sharp cost of coordinate restrictions

Theorem 3.1 (Shape penalty and dimension threshold). *For every $\eta > 0$ and every $n \geq 3$,*

$$\sup_{\substack{p,q\in\Delta_n^\circ \\ d_{\mathrm{DP}}(p,q)=\eta}} \frac{\Lambda_{\mathrm{mon}}(p,q)}{\Lambda(p,q)} = \sup_{\substack{p,q\in\Delta_n^\circ \\ d_{\mathrm{DP}}(p,q)=\eta}} \frac{\Lambda_{\mathrm{box}}(p,q)}{\Lambda(p,q)} = \frac{2\log((e^\eta+1)/2)}{\eta}. \tag{11}$$

Each supremum is attained by strictly positive endpoints in every such dimension. For $n = 2$ both ratios equal one for all distinct endpoints. The factor in (11) tends to two as $\eta \to \infty$.

Proof. Put $E = e^\eta$ and $B_\eta = 2\log((E+1)/2)$. The affine curve $\ell_i(t) = p_i + t(q_i - p_i)$ has speed

$$D(t) = \max_i \frac{|q_i - p_i|}{\ell_i(t)}.$$

For a ratio $\rho = q_i/p_i \geq 1$ the corresponding term is $(\rho-1)/(1+t(\rho-1))$, increasing in ρ, so it is at most $f(t) = (E-1)/(1+t(E-1))$. For $\rho \leq 1$ the term is at most $g(t) = (1-E^{-1})/(1-t(1-E^{-1}))$, by the same monotonicity in $1-\rho$. These envelopes cross at $1/2$, with f larger on the first half. Hence the affine length satisfies

$$A := \int_0^1 D(t)\,dt \leq \int_0^{1/2} f(t)\,dt + \int_{1/2}^1 g(t)\,dt = B_\eta. \tag{12}$$

Since $p \neq q$, the continuous function D is positive and bounded away from zero. The map $s(t) = A^{-1}\int_0^t D(u)\,du$ therefore has a continuously differentiable inverse. Reparameterizing ℓ by s gives cost exactly A and preserves coordinate monotonicity and endpoint ranges. Because $\Lambda \geq d_{\mathrm{DP}}(p,q) = \eta$, both ratios are at most B_η/η.

For equality choose $0 < a \leq 1/(2E)$, $b = aE$ and $c = 1-a-b > 0$. Theorem 2.1 gives $\Lambda = \eta$, since $b \leq 1/2$. Both restricted classes force the reservoir to stay at c. The remaining coordinates have constant sum $m = a + b$; dividing them by m preserves their log derivatives. Their normalized endpoints are $(1/(E+1), E/(E+1))$ and its swap. Equation (8) gives restricted optimum B_η, attained by a monotone binary path. Thus both upper bounds are attained in dimension three.

For $n > 3$, split c into $n-2$ positive masses. Split the reservoir of the unrestricted path in these fixed proportions; its log speeds are preserved. Conversely, aggregating these coordinates preserves the derivative bound, so the unrestricted optimum remains η. Every restricted path fixes each split coordinate, giving the same binary restricted optimum. For $n = 2$, the unrestricted binary optimum in (8) already satisfies both restrictions. Finally $B_\eta/\eta \to 2$ follows directly from its formula. □

The comparison is with the actual unrestricted optimum. In particular, in the attaining triplets the endpoint distance itself is achievable, while preserving endpoint ranges requires the larger budget B_η. A coordinate with equal endpoints is enough to create the obstruction once three outputs are available.

4 Exact necessary certificates for general endpoints

The same endpoint envelopes give a certificate for the peak of any selected coordinate, without constructing a path. Write $Z = e^L$; rational input to the algorithm means rational p, q, Z, even when L is not rational.

Theorem 4.1 (Excursion-certificate sweep). *Let $n \geq 2$, $p, q \in \Delta_n^\circ$ and $Z = e^L \geq \max_i\{p_i/q_i, q_i/p_i\}$. For each coordinate j, define*

$$S_j(z) = \sum_{i\neq j} \min\{p_i z, Zq_i/z\}, \quad 1 \leq z \leq Z, \qquad C_j = 1 - \min_{1\leq z\leq Z} S_j(z). \tag{13}$$

Every path of budget at most L satisfies

$$\max_t r_j(t) \geq C_j \geq \max\{p_j, q_j\}.$$

For rational p, q, Z, one can compute $C_j = 1 - \sqrt{R_j}$ with R_j rational in $O(n\log n)$ rational arithmetic operations and comparisons, plus one square root. All coordinates take $O(n^2 + n\log n)$ such operations and at most n

square roots. These are arithmetic-operation bounds. For the reservoir of Theorem 2.1, the certificate equals the true minimum peak at every feasible budget.

Proof. Normalization and (5) give $r_j(t) \geq 1 - S_j(e^{Lt})$. Taking the maximum proves the certificate for $L > 0$. If $L = 0$, the endpoint hypothesis forces $p = q$ and every admissible path is constant, giving the same formula directly. At $z = 1$ every minimum selects p_i, and at $z = Z$ it selects q_i, proving endpoint inclusion. If the endpoint hypothesis fails, the endpoint distance already rules out the budget.

Put $v = z^2$. Coordinate i changes branch at the rational breakpoint $v_i = Zq_i/p_i \in [1, Z^2]$. Sort q_i/p_i and group equal values. On any cell the envelope sum is $Az + B/z$, where $A, B \geq 0$ are rational. Initially $A = 1 - p_j$, $B = 0$. At each event, transfer every $i \neq j$ in the group by

$$A \leftarrow A - p_i, \qquad B \leftarrow B + Zq_i.$$

For a closed cell $v \in [\ell, h]$, its minimum squared value is the smallest of

$$\frac{(A\ell + B)^2}{\ell}, \qquad \frac{(Ah + B)^2}{h}, \qquad 4AB \ \text{ if } A, B > 0 \text{ and } \ell \leq B/A \leq h. \tag{14}$$

Indeed $Az + B/z$ has its only possible interior minimum at $z^2 = B/A$; if either coefficient vanishes it is monotone. It is nonnegative, so squaring preserves order. The minimum R_j of these rational candidates gives exactly $C_j = 1 - \sqrt{R_j}$.

There are $O(n)$ grouped events after sorting, with constant work per transferred index and cell. Reuse the common sorted order for every j, allowing the harmless event of j itself and skipping its transfer. A linear scan per coordinate proves the all-coordinate bound. At $Z = 1$ the sole interval is degenerate and $R_j = (1 - p_j)^2$. The operation count does not bound bit complexity.

For the swapped reservoir the two events are Za/b and Zb/a, inside $[1, Z^2]$ because $Z \geq e^{T_*} \geq b/a$. On the first, middle and last cells, respectively, the sum is

$$(a + b)z, \qquad az + Za/z, \qquad Z(a + b)/z.$$

The middle stationary point $z = \sqrt{Z}$ lies in its cell. The exterior minima equal $a + b$, so $\min S_0 = \min\{a + b, 2a\sqrt{Z}\}$. Therefore $C_0 = \max\{c, 1 - 2a\sqrt{Z}\}$, attained by Theorem 2.1. □

The certificate supplies a necessary lower bound for arbitrary endpoints. Its exact evaluation does not establish general cap sufficiency or solve general path feasibility.

Reproducible checks. A standard-library Python implementation accompanies the sweep, together with serialized piecewise paths for 657 small-b and 210 large-b instances. A separate verifier checks positivity, normalization, joins, derivative bounds and caps on complete continuous cells by quadratic inequalities. Across the 867 paths it checks 1,824 half-cells and 5,472 coordinate cells; reflection certifies the second halves. A further comparison reconstructs envelope cells directly for 233 endpoint pairs in dimensions two through eight and three budgets per pair, giving 3,483 coordinate comparisons. These finite checks support the implementations; the quantified results follow from the proofs above.

Remaining questions. The original general interpolation question, including a complete comparison of normalized geometric and transform-linear families, remains unresolved by these subsidiary results. In particular, a monotone scalar construction before normalization need not satisfy (2) afterward. General cap-feasibility criteria, a formula for the shape-restricted optimum at arbitrary endpoints, and sharper bit-complexity bounds for excursion certificates are natural further questions.

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References

[BCSZ18] Christian Borgs, Jennifer Chayes, Adam Smith, and Ilias Zadik. Private algorithms can always be extended. arXiv:1810.12518v2, 2018.

[DRS19] Jinshuo Dong, Aaron Roth, and Weijie J. Su. Gaussian differential privacy. arXiv:1905.02383v3, 2019.

[FHMRŻ26] Alfonso Fernández de Bobadilla, Mykhailo Hontarenko, Guillem Müller-Rigat, and Karol Życzkowski. Reachable sets under Kolmogorov dynamics in the probability simplex. arXiv:2609.38382v1, 2026.

[LQ26] Ruiyao Liu and Chenxi Qiu. A graph-based framework for extending metric differential privacy mechanisms. arXiv:2609.14125v1, 2026.

[LR15] Bas Lemmens and Mark Roelands. Unique geodesics for Thompson's metric. *Annales de l'Institut Fourier*, 65(1):315–348, 2015. doi:10.5802/aif.2932.

[Qiu26] Chenxi Qiu. Interpolation-based optimization for enforcing ℓ_p-norm metric differential privacy in continuous and fine-grained domains. arXiv:2601.09946v1, 2026.